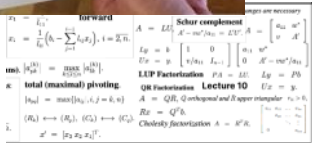


$$3) A = \begin{bmatrix} 2 & 4 & 2 \\ 4 & -10 & 2 \\ 1 & 2 & 4 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 4 & -10 & 2 \\ 2 & 4 & 2 \\ 1 & 2 & 4 \end{bmatrix} \xrightarrow{\begin{matrix} R_2 \leftarrow R_2 - \frac{1}{2}R_1 \\ R_3 \leftarrow R_3 - \frac{1}{2}R_1 \end{matrix}} \begin{bmatrix} 4 & -10 & 2 \\ 0 & 9 & 1 \\ 0 & 0 & 3 \end{bmatrix}$$



$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{4} & \frac{1}{2} & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 4 & -10 & 2 \\ 0 & 9 & 1 \\ 0 & 0 & 3 \end{bmatrix}$$

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



2. Solving the system $Ax = b$ is now equivalent to solving two triangular systems

$$Ly = Pb \text{ and } Ux = y$$

$$Ax = \begin{bmatrix} 6 \\ -10 \\ 9 \end{bmatrix} \xrightarrow{L \triangleright b} Ly = Pb$$

$$Ly = Pb \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{4} & \frac{1}{2} & 1 \end{bmatrix} y = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 6 \\ -10 \\ 9 \end{bmatrix}$$

$$\begin{cases} y_1 = -10 \\ \frac{1}{2}y_1 + y_2 = 6 \\ \frac{1}{4}y_1 + \frac{1}{2}y_2 + y_3 = 9 \end{cases}$$

$$y_2 = 11$$

$$-\frac{5}{2} + \frac{11}{2} + y_3 = 9 \Rightarrow$$

$$\Rightarrow -5 + 11 + 2y_3 = 18 \Rightarrow 2y_3 = 12 \Rightarrow y_3 = 6$$

$$y = \begin{bmatrix} -10 \\ 11 \\ 6 \end{bmatrix}$$

$$Ux = y$$

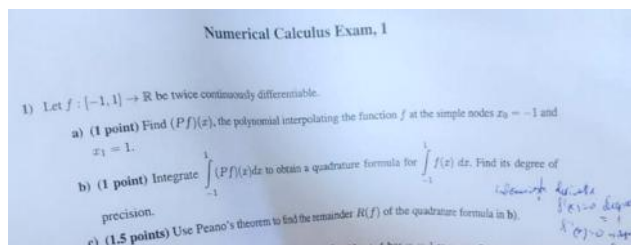
$$\begin{bmatrix} 4 & -10 & 2 \\ 0 & 9 & 1 \\ 0 & 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -10 \\ 11 \\ 6 \end{bmatrix} \rightarrow \begin{cases} x_3 = 2 \\ x_2 = 1 \\ x_1 = -1 \end{cases}$$

$$\begin{bmatrix} 4 & -10 & 2 \\ 0 & 9 & 1 \\ 0 & 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -10 \\ 11 \\ 6 \end{bmatrix}$$

$$\rightarrow \begin{cases} 4x_1 - 10x_2 + 2x_3 = -10 \\ 9x_2 + x_3 = 11 \\ 3x_3 = 6 \end{cases}$$

$$x_1 = -1$$

$$x = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$



$$x_0 = -1, x_1 = 1 \quad L_n f(x) = \sum_{i=0}^n e_i(x) f(x_i) \quad e_i$$

$$e_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x - 1}{-1 - 1} = \frac{1}{2}(1 - x)$$

$$e_1(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x + 1}{1 - (-1)} = \frac{1}{2}(1 + x)$$

$$\begin{aligned} L_n f(x) &= \frac{1}{2}(1 - x)f(-1) + \frac{1}{2}(1 + x)f(1) \\ &= \frac{1}{2}(f(-1) - xf(-1) + f(1) + xf(1)) \\ &= \frac{1}{2}(x(f(1) - f(-1)) + f(1) + f(-1)) \\ &= \frac{1}{2}x(f(1) - f(-1)) + \frac{1}{2}(f(1) + f(-1)) \end{aligned}$$

$$b) \int_{-1}^1 (Pf)(x) dx = \int_{-1}^1 f(x) dx$$

$$\begin{aligned} \int_{-1}^1 L_n f(x) dx &= \frac{1}{4}(f(1) - f(-1))x^2 + \frac{1}{2}(f(-1) + f(1))x \\ &= \frac{1}{4}(f(1) - f(-1)) \cdot 1^2 + \frac{1}{2}(f(-1) + f(1)) \cdot 1 - \\ &\quad - \frac{1}{4}(f(1) - f(-1)) \cdot (-1)^2 - \frac{1}{2}(f(-1) + f(1)) \cdot (-1) \\ &= f(-1) + f(1) \end{aligned}$$

$$p(x) = \prod_{\substack{j=0 \\ j \neq i}}^m \frac{(x - x_j)}{(x_i - x_j)}$$

$$x / -1$$